

Improving Wireless Network Security Based On Radio Fingerprinting

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Abstract—With the rapid development of the popularity of wireless networks, there are also increasing security threats that follow, and wireless network security issues are becoming increasingly important. Radio frequency fingerprints generated by device tolerance in wireless device transmitters have physical characteristics that are difficult to clone, and can be used for identity authentication of wireless devices. In this paper, we propose a radio frequency fingerprint extraction method based on fractional Fourier transform for transient signals. After getting the features of the signal, we use RPCA to reduce the dimension of the features, and then use KNN to classify them. The results show that when the SNR is 20dB, the recognition rate of this method is close to 100%.

Index Terms—Wireless Network Security; Radio Frequency Fingerprints; Transient Signals

I. INTRODUCTION

With the development of wireless communication and network technology, wireless networks have penetrated various sectors of the national economy such as defense military, teaching research, medical health care, and have been closely related to people's daily lives. Nevertheless, the security of wireless networks is still one of the factors hindering its popularity, mainly in the following aspects[1]. The first one is the openness of the wireless network makes it more vulnerable to malicious attacks. The second one is the mobility of wireless networks makes it difficult to manage security. The third one is the dynamic topology of the wireless network makes the implementation of the security scheme difficult.

In summary, the security problem of wireless networks is caused by the openness of communication carriers, the mobility of node terminals, and the dynamically changing network topology. The security threats to wireless network node terminals are mainly: the attackers pretend to be a legitimate user, illegal access to network resources, implement a denial of service attack, connecting to the attacking object network through the wireless network to implement an attack, obtaining management control rights to the attack target network through the wireless network [2]

The actual wireless network security mechanism often has defects. For example, the IEEE 802.11 wireless local area network (WLAN) security mechanism evolved from IEEE 802.1X authentication to IEEE 802.11i from the original Wired Equivalent Privacy (WEP). However, this security mechanism still does not guarantee that there are no security issues.

Besides, if the key is compromised, the existing security mechanism simply cannot achieve its claimed authentication service [3]. Therefore, the current security mechanisms running on the wireless network link layer and above need to be further enhanced. The security mechanism running on the physical layer of the wireless network has been studied by many scholars. The security mechanism running at the physical layer is simply a hardware-based security mechanism. The current research mainly divides this security mechanism into three categories. The first one is the security mechanism based on circuit delay [4]. The method utilizes inevitable delays within different wireless devices for identification. Due to individual differences in each wireless device, the delay in operation is unique, and the wireless device is securely authenticated according to the delay of the device. The disadvantage of this approach is the need to add additional custom hardware to existing wireless devices. The second one is the security mechanism based on clock offset [5]. The basis of this method is that the clocks of different wireless devices have a certain offset. The disadvantage of this method is that an attacker can record the signal of the wireless device, obtain the clock offset information of the wireless device, and then forge the clock offset to attack. The third one is the security mechanism based on "Radio Fingerprinting" [6]-[8]. The method is to confirm the wireless device according to the radio frequency fingerprint of the wireless device, thereby improving the security of the wireless network. The basis of this mechanism is that any wireless device, even the same model, the same batch of wireless devices, will have individual differences and is difficult to replicate. This mechanism works at the physical layer and has higher security than traditional security mechanisms.

"Radio Fingerprint" and "Radio Fingerprint Identification" are new concepts proposed by Hall et al in the study of wireless network device identification such as Bluetooth in 2003 [9]. Since then, the concept has mainly appeared in the identification research of wireless network devices.

The earliest RF fingerprinting technology was proposed in 1995 by Toonstra et al [10]. The feature extraction method based on transient signals was also developed. Most of the early researches focused on transient signal RF fingerprinting. In the on/off transient of the device, the transmitter starts to work. Correspondingly, when the power reaches zero from the rated power, a signal is sent. The process of extracting the

RF fingerprinting of the signal from this part is the transient signal RF fingerprint technology. The generation of such a transient signal is only related to the hardware device and is a signal generated by all wireless communication devices. There is no data information, and it can be used as an independent signal for subsequent analysis and processing. The detection and separation of on/off transients relies on device hardware and channel noise and has proven to be critical to communication systems [11]-[13]. In this paper, we propose a radio frequency fingerprint extraction method based on fractional Fourier transform for transient signals.

In recent years, Fractional Fourier Transform (FrFT) is a new time-frequency analysis tool, which is a broad sense form of Fourier transform[14]-[15]. The representation of the signal on the FrFT domain, while merging the information of the signal in the time and the frequency domain. In recent years, the research on the classification and recognition of the fractional Fourier transform has received extensive attention and is also a new research topic. Fig.1 shows that the overall framework of the paper.

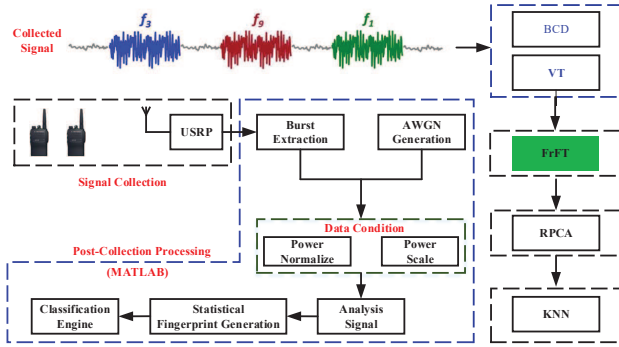


Fig. 1. Wireless device identification process.

In this paper, 10 Motorola walkie-talkies are used to verify the performance of the identification process. As follows, The remaining part of this paper is organized. Section II gives an overview of the methods we used, including FrFT, KNN. Section III shows that the experiment results and the conclusion of this paper is given in Section IV.

II. DEMONSTRATION METHODOLOGY

A. Fingerprint Generation

FrFT can be interpreted as a representation of the FrFT domain formed by the signal rotating in the counterclockwise direction of the coordinate axis around the origin in the time-frequency plane. It is a generalized form of the Fourier transform.

The FrFT of the signal $x(t)$ is defined as:

$$X_\alpha(u) = \{F^\alpha[x(t)]\}(u) = \int_{-\infty}^{\infty} x(t) K_\alpha(t, u) dt \quad (1)$$

The transform kernel $K_\alpha(t, u)$ of the FrFT is:

$$K_\alpha(t, u) = \begin{cases} \delta(t - u) \alpha = 2n\pi \\ \delta(t + u) \alpha = (2n \pm 1)\pi \end{cases} \quad (2)$$

When $\alpha \neq n\pi$,

$$K_\alpha(t, u) = \sqrt{\frac{1-j\cot\alpha}{2\pi}} \exp\left(j\frac{t^2+u^2}{2}\cot\alpha - tu\csc\alpha\right) \quad (3)$$

Where $\alpha = p\pi/2$ is the angle of rotation of the FrFT.

After discretization, FrFT can also be calculated digitally. The most commonly used algorithm at present is the decomposition-type fast algorithm proposed by Ozaktas, which can decompose FrFT into a convolutional form of the signal, and the calculation result is similar to the output of continuous FrFT. The decomposed FrFT transform is:

$$F_p = DK_pJ \quad (4)$$

Where, D and K are the matrix of the interpolation and decimation operations twice, and J is the transformation kernel matrix of the discrete FrFT, namely:

$$K_p = \frac{A_\alpha}{2\Delta x} \exp\left(\frac{j\pi(\cot\alpha)m^2}{(2\Delta x)^2} - \frac{j\pi(\csc\alpha)mn}{(2\Delta x)^2} + \frac{j\pi(\cot\alpha)n^2}{(2\Delta x)^2}\right) \quad (5)$$

The dimension normalized FrFT is defined as:

$$X_p(u) = \sqrt{\frac{1-j\cot\alpha}{2\pi}} \exp[j\pi\cot(\alpha)u^2] \times \int_{-\infty}^{\infty} x(t) \exp[j\pi\cot(\alpha)t^2] \exp[j\pi\cot(\alpha)tu] dt \quad (6)$$

After the signal is subjected to the above FrFT transform, information such as mean, variance, skewness, kurtosis and the like are extracted as fingerprint features of the signal, and the next fingerprint recognition process is performed.

The data is calculated as follows:

The mean is defined as:

$$\mu = \frac{1}{N} \sum_{i=1}^N A_i \quad (7)$$

The variance is defined as

$$V = \frac{1}{N-1} \sum_{i=1}^N |A_i - \mu|^2 \quad (8)$$

The skewness of a distribution is defined as:

$$s = \frac{E(x - \mu)^3}{\sigma^3} \quad (9)$$

The kurtosis of a distribution is defined as:

$$k = \frac{E(x - \mu)^4}{\sigma^4} \quad (10)$$

B. Robust Principal Component Analysis

In the process of identifying the radiation source device, the collected radio frequency signals themselves are "features" that can be distinguished from each other, but because of the high information dimension, it is not convenient to perform fast device identification processing. Moreover, among the original signals or the extracted high-dimensional features, there is a large amount of redundant information, and each feature has a certain correlation, which brings unnecessary burden to subsequent processing. Therefore, it is necessary to reduce the dimension of high-dimensional information to reduce the feature dimension and correlation, but this will cause a certain loss of information on the other hand, and introduce errors into the device. Therefore, the most useful information is retained while reducing the dimension. Primary purpose. Robust Principle Component Analysis (RPCA) is an excellent dimensionality reduction method that can achieve this goal.

RPCA can recover the useful low-rank matrix in the data and eliminate the sparse matrix of noise under certain conditions.

The specific problem solved by RPCA is to decompose the original data matrix $\mathbf{X} = \mathbf{R}^{l \times n}$ into a low-rank matrix $\mathbf{L}$ and a sparse matrix $\mathbf{S}$, which is mathematically expressed as a convex optimization problem:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{S}} & \|\mathbf{L}\|_* + \lambda \|\mathbf{S}\|_1 \\ \text{s.t.} & \mathbf{X} = \mathbf{L} + \mathbf{S} \end{aligned} \quad (11)$$

Among them, $\|\cdot\|_*$ and $\|\cdot\|_1$ represent the kernel norm and l_1 norm of the solution matrix respectively. $\lambda > 0$ is the balance factor, which is used to adjust the ratio between the low-rank matrix and the sparse matrix. The k value can be adjusted according to the specific problem involved to adjust $\lambda_k = k/\sqrt{\max(l, n)}$ and $\lambda = 1/\sqrt{\max(l, n)}$ are usually taken.

C. Device Identification

In this paper, Classification performance is evaluated using the K-Nearest Neighbor (KNN). KNN (K-Nearest Neighbor) is one of the simplest machine learning algorithms that can be used for classification and regression. It is a supervised learning algorithm. The idea is that if a sample has most of the K's most similar samples in the feature space (ie, the nearest neighbor in the feature space) belongs to a certain category, then the sample also belongs to this category. That is to say, the method determines the category to be divided into samples according to the category of the nearest one or several samples in the class decision.

III. EXPERIMENT AND RESULTS

A. Signal Collection and Identification System

Figure 1 displays the realization of signal collection and data processing. Signal collection, data conditioning, fingerprint generation, and the device identification formed the identification system together with a receiver and a computational platform. All the signals are collected with USRP2940 with

subsequent processing performed by the computational platform..

B. Classification Performance

For getting better classification performance and reducing the calculation, the RPCA method is been used to reduce the dimension of the feature. The features after dimensionality reduction can be used to replace the original features to perform the classification.

For a better classification performance, the Kfold cross validation is used for evaluation. Consistent with common practice is used such that samples of each device are divided into five blocks, among which four blocks of the device are used to train and the rest one to test.

As shown in Fig. 2 is the FrFT-based feature extraction method. When the SNR is 20dB, the obtained average recognition rate varies with the fractional order. It can be seen that the best classification identification is obtained when the order is 0.8. The effect, so this simulation is performed at 0.8 order.

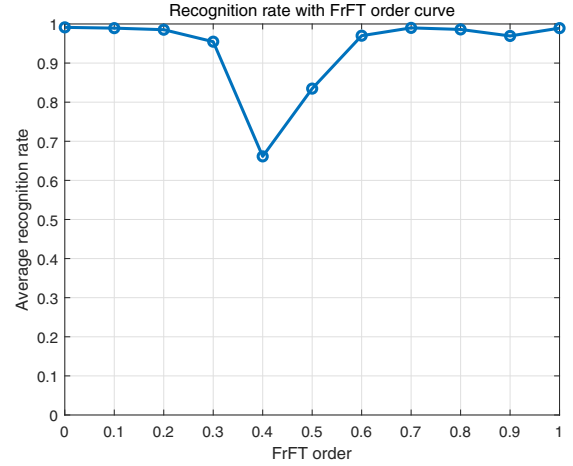


Fig. 2. Recognition rate as a function of order when SNR is 20dB.

Fig 3 shows the feature dimensionality reduction visualization image with SNR of 0 dB in the case of 0.8-order FrFT.

As shown in Fig. 4, the confusion matrix is obtained when the SNR is 0 dB in the case of 0.8-order FrFT.

Fig. 5 shows the characteristic dimensionality reduction visualization image with SNR of 20 dB in the case of 0.8-order FrFT. As shown in Fig. 6, the characteristic dimensionality reduction confusion matrix is obtained when the SNR is 20 dB in the case of 0.8-order FrFT. With the signal collection and identification system is shown in fig. 1, the identification rate curve under different SNR of 0 20dB is shown in Fig 7. Fig. 7 shows the classification and recognition of the FrFT feature extraction based on order 0.8, which varies with the signal-to-noise ratio. It can be seen that as the signal-to-noise ratio increases, the recognition rate also increases.

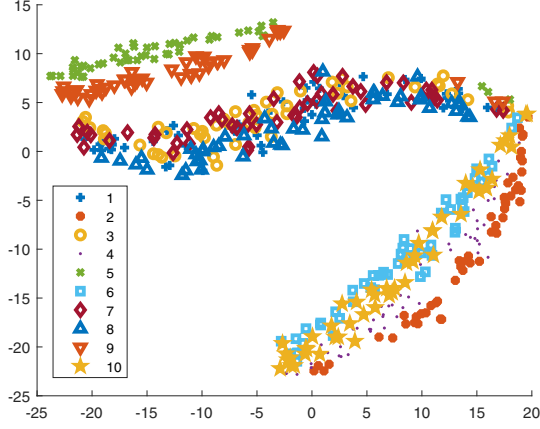


Fig. 3. Reduced-dimensional visualization of SNR of 0 dB at 0.8-order FrFT and the dimensional is 2.

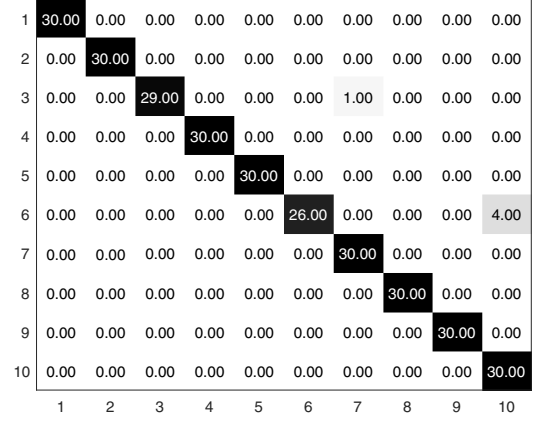


Fig. 6. Confusion matrix with SNR of 20dB at 0.8-order FrFT.

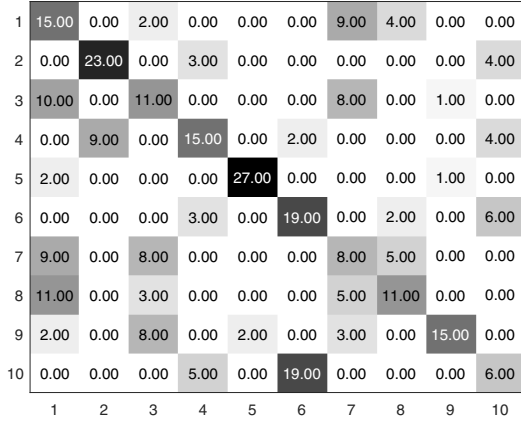


Fig. 4. Confusion matrix with SNR of 0 dB at 0.8-order FrFT.

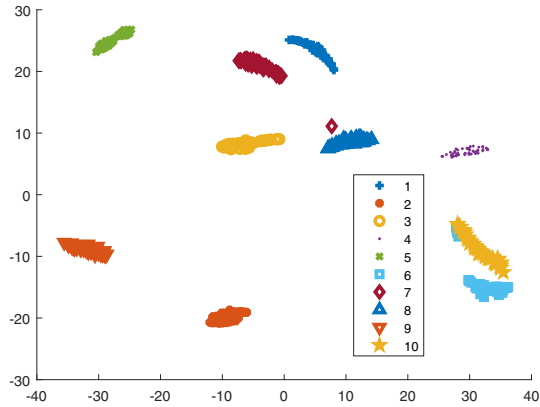


Fig. 5. Reduced-dimensional visualization with SNR of 20dB at 0.8-order FrFT and the dimensional is 2.

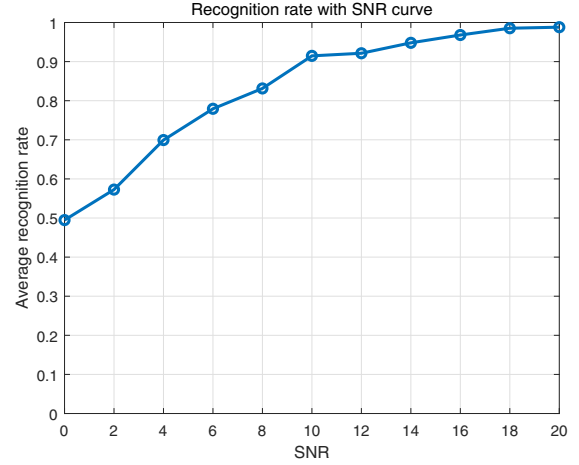


Fig. 7. The recognition rate changes with SNR when the order is 0.8..

IV. CONCLUSION

In this paper, we present a method based on radio frequency fingerprint to enhance wireless network security. We use Fractional Fourier Transform to represent the signal on the fractional Fourier domain, after getting the fractional Fourier domain representation, we divide the signal data into 20 segments and calculate the mean, variance, skewness, and kurtosis of each segment as the radio frequency fingerprint to distinguish different wireless network devices. We use RPCA to reduce the dimension of features and use KNN to classify. In this paper, 10 Motorola walkie-talkies are used to verify the performance of the identification process. The results show that when the SNR is 20dB, the recognition rate is close to 100%.

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